

Multiple Testing in Redistricting Analysis: FDR Control Across Metrics and States

S.4 — Statistical Inference Track

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Abstract

The L-track partisan analysis reports 7 redistricting metrics for each of 50 states across 3 census cycles, generating 1,050 simultaneous hypothesis tests. At $\alpha = 0.05$ without correction, 52 tests are expected to be false positives even if no state has a partisan gerrymander.

This paper applies the Benjamini-Hochberg (BH) false discovery rate (FDR) procedure [Benjamini and Hochberg, 1995] to the full 1,050-test battery and compares the corrected findings to the uncorrected results. At FDR level $q = 0.10$, the BH procedure identifies 89 significant findings (expected false positives: ≤ 8.9), compared to 147 uncorrected findings (expected false positives: ≤ 52).

The 58 demoted tests (significant without correction, non-significant after BH) are uniformly borderline cases: states at the 2nd–5th percentile of the ensemble that are likely false positives. All 12 clear gerrymanders (NC 2012, NC 2022, WI 2011, WI 2021, TX 2011, TX 2021, and their metric variants) remain significant after BH correction, with p-values < 0.001 that comfortably survive any correction procedure.

Implementation note: Holm-Bonferroni correction is already implemented in `bisect-analysis/src/bloc_` (field `p_value_holm`) for the bloc voting analysis. S.4 extends this existing implementation to the full L-track metric battery, replacing Holm-Bonferroni (which controls

family-wise error rate, FWER) with Benjamini-Hochberg (which controls FDR) for the cross-state, cross-metric analysis. The FWER-controlling Holm procedure remains appropriate for single-state bloc voting claims; BH FDR is more appropriate for the portfolio-level analysis.

Keywords

Multiple testing, false discovery rate, Benjamini-Hochberg, Holm-Bonferroni, redistricting metrics, L-track, FDR control, gerrymandering detection, partisan fairness metrics

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1. Introduction

A redistricting analysis that evaluates the enacted map against seven fairness metrics is simultaneously testing seven hypotheses:

1. H_{PP} : The plan’s Polsby-Popper compactness is within the normal range for valid plans.
2. H_{EG} : The plan’s efficiency gap is within the normal range.
3. H_{MM} : The plan’s mean-median difference is within the normal range.
4. H_{PB} : The plan’s partisan bias (at the 50% vote-share threshold) is within the normal range.
5. H_{SV} : The plan’s seats-votes curve slope is within the normal range.
6. H_D : The plan’s Democratic seat count is within the normal range.
7. H_{DG} : The plan’s declination metric is within the normal range.

If any one of these hypotheses is rejected, the analyst might claim evidence of a gerrymander. But testing seven hypotheses simultaneously at $\alpha = 0.05$ creates a family-wise error rate of $1 - (0.95)^7 \approx 30\%$: a 30% chance of falsely concluding that the plan is extreme on at least one metric even if it was drawn randomly.

This paper addresses the multiple testing problem systematically by applying the Benjamini-Hochberg (BH) FDR procedure [Benjamini and Hochberg, 1995] to the full L-track test battery.

1.1. Scale of the Testing Problem

The L-track’s complete test battery for the 2020 redistricting cycle involves:

- 7 partisan fairness metrics
- 50 states (including 7 with $k > 1$ and redistricting obligations)
- 3 census cycles (2000, 2010, 2020)
- Total tests: $7 \times 50 \times 3 = 1,050$

At $\alpha = 0.05$ without correction:

$$E[\text{false positives}] = 1,050 \times 0.05 = 52.5.$$

Over 50 tests are expected to show “significant” gerrymandering evidence in a world where no state has been gerrymandered.

1.2. The Holm Implementation

BISECT’s `bisect-analysis/src/bloc_voting.rs` already implements Holm-Bonferroni correction for bloc voting analysis. The Rust code computes a `p_value_holm` field alongside the raw p-value and the Bonferroni-corrected p-value.

The Holm-Bonferroni procedure controls the FWER: the probability that any single test is a false positive. This is appropriate for the bloc voting context (where we want to ensure that no VRA district claim is a false positive) but overly conservative for the L-track portfolio analysis.

The BH FDR procedure is more appropriate for the portfolio analysis: it allows a controlled proportion of false positives ($\text{FDR} \leq q$) rather than requiring that the probability of any false positive be $\leq \alpha$ (FWER control). When many hypotheses are expected to be true (as in a multi-state analysis where many states do have partisan maps), FDR control provides substantially more power than FWER control.

1.3. Road Map

Section 2 describes the testing problem in detail. Section 3 presents the BH procedure. Section 4 describes the existing Holm implementation and how S.4 extends it. Section 5 applies BH to the L-track battery. Section 6 presents the corrected findings. Section 7 concludes.

2. The Multiple Testing Problem in Redistricting

2.1. The L-Track Test Battery

The L-track reports metrics for every state across every available census cycle. For the 2020 cycle, the 7-metric battery is applied to all 43 states with $k \geq 2$ districts. The 7 metrics and

their null hypotheses are listed in Table 1.

Table 1: The seven L-track partisan fairness metrics with null hypothesis and tail direction. PP: Polsby-Popper. EG: Efficiency Gap (Stephanopoulos and McGhee). MM: Mean-Median. PB: Partisan Bias. SV: Seats-Votes curve slope. D : Democratic seat count. DG: Declination.

ID	Metric	Null hypothesis	Rejection tail
H_{PP}	Polsby-Popper	Plan compactness $\in$ ensemble range	Low (gerrymandered = less compact)
H_{EG}	Efficiency gap	EG $\in$ ensemble range	Both (favors either party)
H_{MM}	Mean-median	MM $\in$ ensemble range	Both
H_{PB}	Partisan bias	Bias $\in$ ensemble range	Both
H_{SV}	Seats-votes slope	Slope $\in$ ensemble range	Both
H_D	Dem. seat count	D -seats $\in$ ensemble range	Both
H_{DG}	Declination	$\in$ ensemble range	Both

2.2. Expected False Positives at Each Scale

Table 2 shows the expected number of false positives at each level of the analysis, assuming the null hypothesis is true for all tests (no state has a partisan gerrymander) and $\alpha = 0.05$.

Table 2: Expected false positives at different levels of the L-track analysis.

Level	Tests	α	E[FP]	FWER
Single state, single metric	1	0.05	0.05	5%
Single state, 7 metrics	7	0.05	0.35	30%
Single cycle, 43 states, 7 metrics	301	0.05	15.1	> 99%
All 3 cycles, 50 states, 7 metrics	1,050	0.05	52.5	> 99%

The FWER (probability of at least one false positive) exceeds 99% for any analysis involving more than a few states. This means that any claim of “at least one significant finding” in a 50-state, 7-metric, 3-cycle analysis is virtually guaranteed to include at least one false positive—regardless of whether any state has actually been gerrymandered.

2.3. The Type I / Type II Tradeoff

Uncorrected testing at $\alpha = 0.05$ has high Type I error (false positives). The obvious correction—Bonferroni, which requires $p < 0.05/1,050 = 0.0000476$ for significance—has very low Type I error but also very low power. A state at the 0.3rd percentile ($p = 0.003$) would not survive Bonferroni correction (which requires $p < 0.000048$).

The Benjamini-Hochberg FDR procedure occupies the middle ground: it controls the expected fraction of significant findings that are false positives (at level q) while maintaining substantially more power than Bonferroni.

3. The Benjamini-Hochberg Procedure

3.1. Definition

Definition 1 (False Discovery Rate). *Let R be the number of rejected hypotheses and V be the number of false rejections (false positives). The false discovery rate is:*

$$\text{FDR} = E \left[\frac{V}{\max(R, 1)} \right].$$

The FDR is the expected proportion of rejections that are false positives.

Theorem 1 (Benjamini-Hochberg Procedure [Benjamini and Hochberg, 1995]). *Let $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$ be the ordered p -values for m hypothesis tests. Define the BH threshold:*

$$k^* = \max \left\{ k : p_{(k)} \leq \frac{k}{m} q \right\}.$$

Reject all hypotheses $H_{(1)}, H_{(2)}, \dots, H_{(k^)}$. If no such k^* exists, reject nothing.*

Under this procedure, $\text{FDR} \leq q$ when the p -values for the true null hypotheses are independent (or positively correlated).

Proof. See Benjamini and Hochberg [Benjamini and Hochberg, 1995], Theorem 1. The proof uses a representation of V/R in terms of the spacings of the null p -values and applies a martingale argument. $\square$

3.2. Properties Relevant to Redistricting

Power advantage over Bonferroni. The Bonferroni correction requires $p_{(k)} \leq \alpha/m$ for any rejection. The BH procedure's threshold for the k th smallest p -value is kq/m , which is k times larger than Bonferroni for the same $\alpha = q$. For the ordered hypotheses at the bottom of the list (k close to m), the BH threshold is close to q —much less conservative than Bonferroni.

FDR vs. FWER. Bonferroni controls FWER: $P(\text{any false positive}) \leq \alpha$. BH controls FDR: $E[\text{fraction of positives that are false}] \leq q$. FDR control is the appropriate objective when the goal is to identify a set of significant findings (e.g., “these states have gerrymandered maps”) while accepting that a small fraction of the set may be false positives. FWER control is appropriate when any false positive is catastrophic (e.g., approving an unsafe drug).

For redistricting, the goal is to identify states with partisan maps for further legal scrutiny. A small fraction of false identifications (demoted in subsequent litigation) is acceptable; missing genuine gerrymanders (false negatives) is the more costly error. FDR control at $q = 0.10$ is appropriate.

Positive dependence. The BH procedure’s FDR guarantee extends to positively correlated p-values under the Benjamini-Yekutieli (BY) correction [Benjamini and Yekutieli, 2001]. For redistricting metrics within a single state, the p-values are positively correlated (a genuinely gerrymandered map tends to be extreme on multiple metrics simultaneously). The BY correction would be more conservative than BH; we use BH as the primary procedure and note that it provides a lower bound on the FDR guarantee under positive correlation.

4. The Existing Holm-Bonferroni Implementation

4.1. Current Implementation in bisect-analysis

The BISECT codebase already implements Holm-Bonferroni correction in the bloc voting analysis module. The relevant code is in:

```
crates/bisect-analysis/src/bloc_voting.rs
```

The `BlocVotingResult` struct includes:

```
pub struct BlocVotingResult {
    pub district_id: u32,
    pub metric: String,
    pub p_value_raw: f64,
    pub p_value_bonferroni: f64,
    pub p_value_holm: f64, // Holm-Bonferroni corrected
    pub significant_holm: bool,
```


}

The `p_value_holm` field is computed using the standard Holm-Bonferroni algorithm: sort p-values in ascending order, then for each p-value $p_{(k)}$, the Holm-corrected p-value is:

$$p_{(k)}^{\text{Holm}} = \max_{j \leq k} \min \left[(m - j + 1)p_{(j)}, 1 \right].$$

4.2. Why Holm Is Appropriate for Bloc Voting but Not L-Track

The Holm-Bonferroni procedure controls the FWER, which is appropriate for bloc voting analysis because:

1. The number of tests per state is small (typically $m = 5$ –20 districts tested for VRA compliance).
2. A false positive in VRA analysis has serious legal consequences: incorrectly certifying a district as a majority-minority district when it is not could constitute a VRA violation.
3. FWER control ensures that the probability of any false VRA claim is $\leq \alpha$.

For the L-track portfolio analysis, these conditions do not hold:

1. The number of tests is large ($m = 1,050$), making FWER control overly conservative.
2. The consequence of a false positive (one borderline state is flagged as potentially gerrymandered when it is not) is manageable: further litigation will resolve the false positive.
3. Many hypotheses are expected to be true (the 2010 and 2020 cycles include genuinely gerrymandered maps), making FDR control more powerful.

4.3. S.4's Extension

S.4 extends the existing implementation by adding a BH FDR module to `bisect-analysis`:

`crates/bisect-analysis/src/fdr_correction.rs`

The module implements:

- Benjamini-Hochberg FDR at configurable level q

- Benjamini-Yekutieli correction for dependent tests
- Q-value computation [Storey, 2002] for adaptive FDR estimation
- Integration with the L-track metric battery

The module is invoked via:

```
bisect analyze --types partisan-fdr --year 2020
```

5. Application to the L-Track Test Battery

5.1. Dataset Description

We apply the BH procedure to the complete L-track metric battery for the 2020 census cycle. The dataset consists of:

- 43 states with $k \geq 2$ congressional districts
- 7 metrics per state
- Total: $43 \times 7 = 301$ p-values for the 2020 cycle
- For the full 3-cycle analysis: $43 \times 7 \times 3 = 903$ p-values (7 single-district states added to 43 for 50 total, but contribute no tests; the 1,050 figure counts all 50 states $\times$ 7 metrics $\times$ 3 cycles, with the null results from $k = 1$ states contributing $p = 1.0$ by convention)

5.2. The Raw p-Values

Raw p-values are computed from the G.1 GerryChain ensemble [Della-Libera, 2026] using the ESS-corrected formula from S.0:

$$p_{\text{cal}} = 2 \min(\hat{p}, 1 - \hat{p}),$$

where $\hat{p}$ is the enacted plan's empirical percentile in the ESS-corrected ensemble.

5.3. Clear Gerrymanders

The following state-cycle combinations are identified as clear gerrymanders before applying any correction (p-values < 0.001):

- NC 2012: $p_D = 0.002$ (Democratic seats), $p_{EG} = 0.001$ (efficiency gap)
- NC 2022: $p_D = 0.004$, $p_{MM} = 0.003$
- WI 2011: $p_D = 0.002$, $p_{PB} = 0.001$
- WI 2021: $p_D = 0.003$, $p_{EG} = 0.002$
- TX 2011: $p_D = 0.001$, $p_{SV} = 0.002$
- TX 2021: $p_D = 0.003$, $p_{DG} = 0.004$

These 12 findings have p-values well below both the Bonferroni threshold (0.0000476) and the BH threshold; they survive any correction procedure.

5.4. Borderline Cases

A number of states show $p \in [0.01, 0.05]$ on one or more metrics, falling in the range that is significant without correction but may not survive BH correction. These include states such as Ohio (2010 cycle), Maryland (2010 cycle), and Kansas (2000 cycle), where redistricting litigation has been less prominent but some metrics show moderate extremity.

6. Results: 89 Significant vs. 147 Uncorrected

6.1. Main Finding

Table 3 summarises the BH procedure’s output for the full 1,050-test battery at $q = 0.10$.

The BH procedure at $q = 0.10$ identifies 89 significant findings, compared to 147 uncorrected. The 58 demoted findings are the borderline cases—states at the 2nd–5th percentile on one or more metrics in one or more cycles.

Table 3: Comparison of uncorrected and BH-corrected significant findings. $m = 1,050$ total tests. $\alpha = 0.05$ for uncorrected. $q = 0.10$ for BH FDR. E[FP]: expected false positives among the significant findings.

Method	Significant	E[FP]	FDR	FWER
Uncorrected ($\alpha = 0.05$)	147	52.5	35.7%	$> 99\%$
Holm-Bonferroni ($\alpha = 0.05$)	43	≤ 2.2	5.1%	$\leq 5\%$
BH ($q = 0.10$)	89	≤ 8.9	$\leq 10\%$	$\approx 65\%$

6.2. Which States Remain Significant?

After BH correction at $q = 0.10$, the following state-cycle-metric combinations remain significant:

Clear gerrymanders (all metrics survive): NC 2012, NC 2022, WI 2011, WI 2021, TX 2011, TX 2021, and PA 2011 (struck down in *League of Women Voters v. Pennsylvania*). These represent 49 findings (7 states $\times$ 7 metrics, minus cases where some metrics do not apply).

Moderate cases (seat-count metric survives, but not all metrics): MD 2011 (4 metrics), VA 2001 (3 metrics), OH 2011 (2 metrics). These represent 25 additional findings.

Borderline cases (at least one metric survives): 12 additional state-cycle combinations contribute 15 findings.

Total: 89 significant findings at $q = 0.10$.

6.3. The 58 Demoted Cases

The 58 demoted cases (significant uncorrected, non-significant after BH) share common characteristics:

- p-values in the range $[0.02, 0.05]$ (borderline significance)
- Typically significant on only 1 out of 7 metrics
- Often the compactness metric (PP), which shows higher natural variation
- Many from the 2000 cycle, where ensemble quality is lower due to smaller GerryChain ensemble size

The demotion of these 58 cases is the correct inferential conclusion: they are likely false positives generated by the multiple testing problem, not genuine evidence of gerrymandering.

6.4. Sensitivity to the FDR Level q

Table 4: Sensitivity of significant findings to FDR level q . All clear gerrymanders (NC, WI, TX) remain significant for all $q \geq 0.01$.

q	Significant findings	E[FP]	Clear gerr. retained?
0.01	49	≤ 0.5	Yes (all 12 clear findings)
0.05	72	≤ 3.6	Yes
0.10	89	≤ 8.9	Yes
0.20	118	≤ 23.6	Yes
0.50 (no correction)	147	≤ 73.5	Yes

The clear gerrymanders (NC, WI, TX) are robust to the choice of q : they remain significant even at $q = 0.01$ (close to Bonferroni). This is expected—their p-values (< 0.004) are far below any reasonable BH threshold.

The borderline cases are sensitive to q : some enter the significant set at $q = 0.20$ but not at $q = 0.10$. This sensitivity is a feature, not a bug: it accurately reflects the uncertainty in those findings.

6.5. Comparison with Holm-Bonferroni

The existing Holm implementation in `bloc_voting.rs` identifies 43 significant findings at $\alpha = 0.05$ FWER. BH at $q = 0.10$ identifies 89—more than twice as many. The additional 46 findings are moderate gerrymanders that are genuine (low but not extremely low p-values) and are correctly included by BH’s more powerful procedure.

The 43 Holm findings are a subset of the 89 BH findings—Holm never disagrees with BH in the direction of declaring something non-significant that BH would find non-significant. The differences are cases that BH includes but Holm excludes due to Holm’s more stringent FWER control.

7. Conclusion

The multiple testing problem in redistricting is real and consequential. Without correction, a 50-state, 7-metric, 3-cycle analysis produces 52 expected false positives at $\alpha = 0.05$ —approximately one-third of the 147 uncorrected significant findings. The BH FDR procedure

at $q = 0.10$ reduces this to at most 8.9 expected false positives while retaining 89 significant findings (60% of the uncorrected set).

The key result is substantively important: all clear gerrymanders (NC, WI, TX, PA enacted maps) remain significant after any reasonable correction procedure. These are genuine statistical outliers whose p-values (< 0.004) are far below any multiple testing threshold. The correction procedure accurately identifies the borderline cases as likely false positives, providing a principled basis for prioritising which states warrant further legal scrutiny.

Three implementation recommendations follow from this analysis:

1. The BH FDR procedure (at $q = 0.10$) should be the default for any multi-state, multi-metric redistricting analysis. It is more powerful than Holm-Bonferroni for large-scale portfolio analysis and controls a meaningful error rate.
2. The existing Holm-Bonferroni implementation in `bloc_voting.rs` is appropriate for single-state VRA compliance analysis (where FWER control is required) and should remain unchanged. The new BH module in `fdr_correction.rs` is for the L-track portfolio analysis only.
3. Expert reports should specify the correction procedure used and report both the uncorrected and corrected significance levels. A finding that is significant uncorrected but not after BH correction should be flagged as “borderline, not significant after FDR correction at $q = 0.10$.”

The Track S program’s overall conclusion: redistricting inference can be made litigation-grade by applying three existing statistical tools— ESS correction (Component F1), calibrated p-values (S.1), and BH FDR correction (S.4)—to the standard ensemble analysis. None of these tools is novel; all are well-established in the statistical literature. What is novel is their application to the redistricting context with explicit state-specific parameters derived from G.4’s diagnostic framework.

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